

Lab 1: Requirements Engineering & UML Use-Case Modelling

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1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse uploaded student resumes (PDF), extract skill tags, and match each profile against the active company eligibility rules.	High	PASS: A valid PDF resume is uploaded and the system extracts skill tags and displays whether the student matches each active rule. FAIL: An active-backlog student is marked eligible.	Core function for automated applicant screening and eligibility matching.
FR-002	Functional	The system shall apply multi-tier CGPA and backlog filters to determine each student's eligibility for every active placement drive.	High	PASS: For a test set covering all configured CGPA tiers and backlog states, the eligibility result exactly matches the configured company rules. FAIL: Any student violating a rule is marked eligible.	Prevents ineligible students from entering a placement process.
FR-003	Functional	The system shall allow the Placement Officer to create and manage multi-round interview queues for eligible applicants.	High	PASS: The officer can add eligible applicants to a round, reorder the queue and move an applicant to the next round. FAIL: An ineligible applicant can be added to a queue.	Supports organised management of company selection rounds.
FR-004	Functional	The system shall allow the Placement Officer to record interview outcomes and update an applicant's selection status after each round.	Medium	PASS: The officer can record Selected, Rejected or Next Round, and the applicant's profile shows the saved status. FAIL: An outcome is saved with no applicant or round linked.	Maintains a traceable record of applicants through the drive.
FR-005	Functional	The system shall process offer letters for selected applicants and make the final offer status available to the applicant.	High	PASS: When an applicant is marked Selected, an offer record can be created and its status is visible to the applicant. FAIL: An offer can be issued to an applicant who is not selected.	Completes the placement pipeline from selection to offer.
NFR-001	Nonfunctional	The system shall encrypt all student academic records and offer documents at rest using AES-256.	High	PASS: Security verification confirms that stored academic records and offer documents are encrypted with AES-256. FAIL: Any protected record is stored unencrypted.	Protects sensitive academic and employment data.
NFR-002	Nonfunctional	The system shall return eligibility-screening results for an uploaded resume within 5 seconds under a simulated load of 100 concurrent screening requests.	High	PASS: In load testing, at least 95% of screening requests complete within 5 seconds. FAIL: Fewer than 95% of requests meet the 5-second target.	Lets the placement cell handle large applicant volumes during drives.

2. Actors and Use Cases

Actors: Student Applicant, Placement Officer, Corporate Recruiter

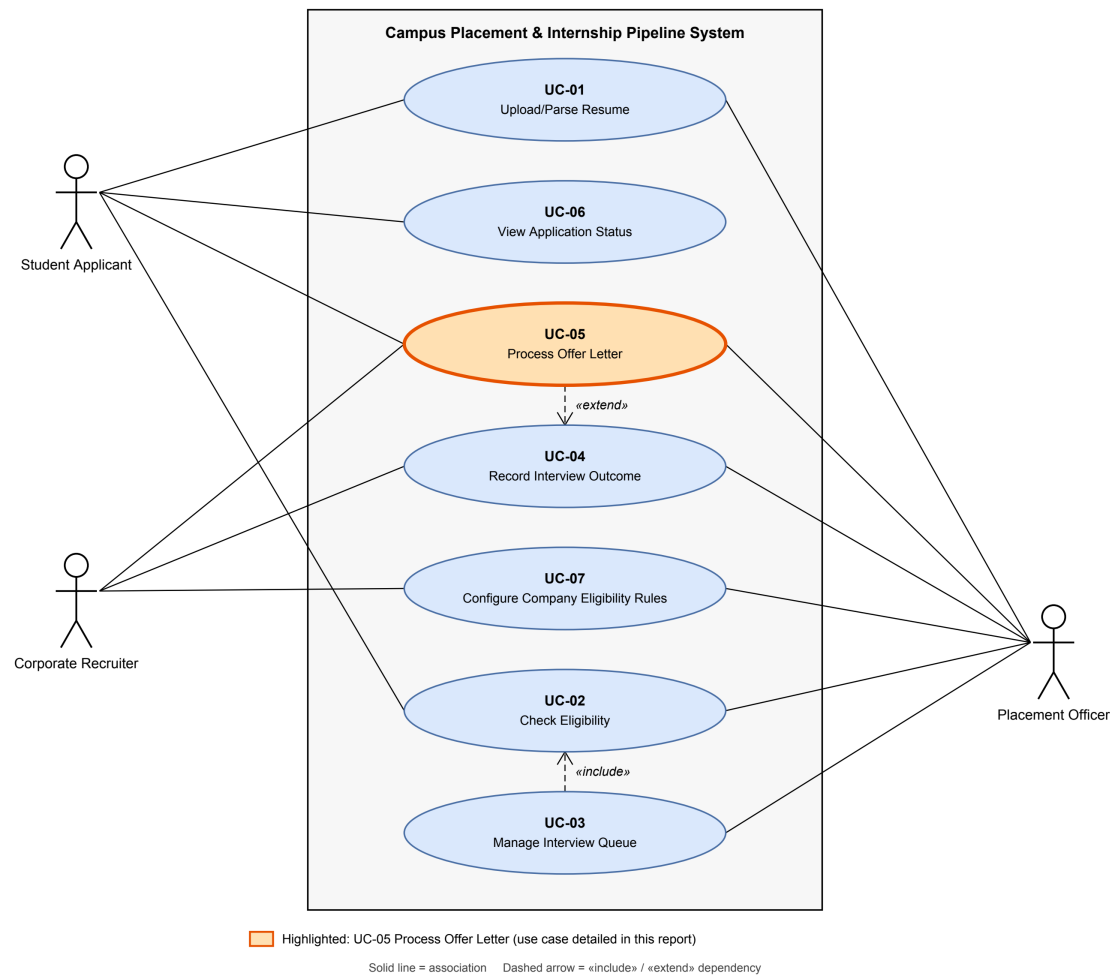
ID	Use Case	Associated Actors
UC-01	Upload/Parse Resume	Student Applicant, Placement Officer
UC-02	Check Eligibility	Placement Officer, Student Applicant
UC-03	Manage Interview Queue	Placement Officer
UC-04	Record Interview Outcome	Placement Officer, Corporate Recruiter
UC-05	Process Offer Letter	Placement Officer, Corporate Recruiter, Student Applicant
UC-06	View Application Status	Student Applicant
UC-07	Configure Company Eligibility Rules	Placement Officer, Corporate Recruiter

Relationships:

- UC-03 Manage Interview Queue «include» UC-02 Check Eligibility (adding an applicant to a queue always checks eligibility).
- UC-05 Process Offer Letter «extend» UC-04 Record Interview Outcome (an offer is processed only when the outcome is Selected).

3. UML Use-Case Diagram

Campus Placement & Internship Pipeline - UML Use-Case Diagram



4. Use-Case Flow - UC-05: Process Offer Letter

Primary Actor	Placement Officer
Supporting Actors	Corporate Recruiter, Student Applicant
Related Requirements	FR-004, FR-005, NFR-001
Preconditions	The applicant has been marked Selected in the final interview round (UC-04).
Postconditions	An offer record is stored (encrypted) and the offer status is visible to the student.
Trigger	The Corporate Recruiter shares the offer details for a selected applicant.

Main Success Scenario

1. Placement Officer opens the list of selected applicants for a drive.
2. System displays applicants whose status is Selected.
3. Placement Officer selects an applicant and chooses "Create Offer".
4. System displays the offer form with the company, role and applicant filled in.
5. Placement Officer enters the CTC and joining date and uploads the offer letter from the recruiter.
6. System verifies that the applicant is Selected and all fields are complete.
7. System saves the offer record in encrypted form with status "Offered".
8. System notifies the student and shows the offer under their applications.
9. Student accepts the offer and the system updates the status to "Accepted".
10. Use case ends successfully.

Alternate Flows

6a. Applicant Not Selected

- 6a1. System detects that the applicant is not marked Selected.
- 6a2. System blocks the offer and displays an error to the Placement Officer.
- 6a3. Use case ends without creating an offer.

9a. Student Declines Offer

- 9a1. Student declines the offer.
- 9a2. System updates the status to "Declined" and notifies the Placement Officer.